

Assessment Tool 2 – Constraint Based Problem Solving

Question 1: Backtracking Search – Doctor Shift Assignment

Problem Statement

Assign three doctors (D1, D2, D3) to three shifts (Morning, Afternoon, Night) while satisfying all given constraints.

Constraints

1. D1 cannot work the Night shift.
2. D2 must work before D3 (Morning < Afternoon < Night).
3. Only one doctor can be assigned to each shift.
4. D3 cannot work the Morning shift.
5. Every doctor must be assigned exactly one shift.

Backtracking Steps

Step 1: Assign D1 → Morning

- Constraint Check:
 - D1 is not assigned Night. ✓
 - No conflict with other doctors. ✓

Step 2: Assign D2 → Afternoon

- Constraint Check:
 - D2 is before D3. ✓
 - Shift is available. ✓

Step 3: Assign D3 → Night

- Constraint Check:
 - D3 is not assigned Morning. ✓
 - Only one doctor per shift. ✓

- All doctors are assigned. ✓

Final Valid Schedule

Doctor	Assigned Shift
D1	Morning
D2	Afternoon
D3	Night

Conclusion

All constraints are satisfied. Therefore, the above assignment is the valid solution obtained using the Backtracking Search algorithm.

Question 2: BFS Search Algorithm

Problem Statement

A robot must travel from the Start (S) position to the Goal (G) in a 5×5 grid while avoiding obstacles.

Constraints

- Move only Up, Down, Left, or Right.
- Cost of each move = 1.
- Obstacles cannot be crossed.
- Manhattan Distance is used as the heuristic.

Step-by-Step BFS Expansion

Initial Queue

Queue = [S]

Step 1

- Expand Start node.
- Queue = [(0,1), (1,0)]

Step 2

- Expand (0,1).
- Queue = [(1,0), (0,2)]

Step 3

- Expand (1,0).
- Queue = [(0,2), (2,0)]

Step 4

- Continue expanding nodes level by level until the Goal node is reached.

Manhattan Distance

Formula:

$$h = |x_1 - x_2| + |y_1 - y_2|$$

Example:

Robot Position = (2,2)

Goal Position = (4,4)

$$h = |2 - 4| + |2 - 4| = 2 + 2 = 4$$

Optimal Path

S → ↓ → ↓ → → ↓ → → ↓ → G

Total Cost

- Number of Moves = **8**
- Cost per Move = **1**

Total Cost = 8

Conclusion

Breadth First Search explores all nodes level by level and guarantees the shortest path when every move has the same cost.

Question 3(a): Constraint Analysis

The robot operates in a partially observable environment where it has limited information.

Constraint Analysis

- The robot can move only in four directions (Up, Down, Left, Right).
- Obstacles block certain paths, requiring alternative routes.
- Every movement costs one unit.
- Risky zones increase the total cost by two additional units.
- The robot can observe only adjacent cells.
- The robot updates its knowledge dynamically as it explores the environment.

Effect on Decision Making

These constraints force the robot to continuously evaluate available paths and choose the safest and least costly route to reach the survivor.

Question 3(b): Solution Approach

Search Strategy: Uniform Cost Search (UCS)

Steps

1. Start from the initial position (S).
2. Observe all adjacent cells.
3. Calculate the movement cost.
4. Avoid blocked cells.
5. Choose the path with the minimum cumulative cost.
6. Update the environment whenever new information is discovered.
7. Continue until the Goal (G) is reached.

Outcome

The robot successfully reaches the survivor while minimizing the overall travel cost.

Question 3(c): Application of AI Concepts

Intelligent Agent

The robot senses its surroundings, processes information, and performs actions to reach the survivor.

Rationality

The robot always selects the action that minimizes total cost while ensuring safe navigation.

Environment Type

- Partially Observable
- Dynamic
- Sequential
- Deterministic

Question 3(d): Justification

Uniform Cost Search is the most appropriate search strategy because:

- It always finds the lowest-cost path.
- It considers both movement cost and risky areas.
- It can adapt when new obstacles are discovered.
- It performs well in partially observable environments.
- It ensures efficient and safe navigation to the goal.

Conclusion

Uniform Cost Search enables the rescue robot to make intelligent decisions, minimize travel cost, and safely reach the survivor despite limited environmental knowledge.